DATA MANAGEMENT CARPENTRY · EPISODE 6

# Storing and Transferring Data Efficiently

Measuring storage, lossless compression techniques, format tradeoffs, and secure transfer channels for collaborative research.

EPISODE SCOPE

# Questions & Objectives

## Core Questions

**How do I measure the size of data?**

**What is compression, and when to use it?**

**When does changing formats help?**

**What are the best transfer methods?**

## Learning Objectives

**Inspect** file and folder sizes in GUI tools.

**Compress** files and folders reliably.

**Differentiate** lossless from lossy algorithms.

**Select** appropriate transfer mechanisms.

NEW DILEMMA

# The Problem: The Data Is Too Big

## Supervisor Request

> *"Can you send the cleaned dataset to our collaborator?"*

The project is organised, but now must be shared across realistic institutional constraints.

## Collaborator Machine Limits

- Older lab PC with very low free drive space
- Constrained, slow campus internet
- Mail server rejects large attachments
- Must reduce size without losing data

BRAINSTORMING EXERCISE

# How Could We Make The Data Smaller?

## Working in pairs, what options can reduce dataset footprint?

- Temporary and cache files?
- File formats (text vs. binary)?
- File compression archives?
- Duplicate records and backups?
- High-resolution raw images & media?
- Intermediate processing stages?

PRACTICAL SOLUTIONS

# Techniques for Reducing Data Size

### 1. File Compression

Bundling files into ZIP/7z archives reduces disk footprint without loss.

### 2. Pruning Scratch Data

Removing temporary caches, duplicated downloads, and redundant copies.

### 3. Efficient Formats

Switching uncompressed raw bitmaps to PNG, or text tables to binary containers.

### Instructor Insight

**Deletion is a last resort!** Better compression and hygiene protect research value.

STORAGE METRICS

# Understanding File Size & Units

STORAGE SCALE

```
KB (Kilobyte)  ≈ 1,000 Bytes
MB (Megabyte)  ≈ 1,000 KB
GB (Gigabyte)  ≈ 1,000 MB  (≈ 1 Billion Bytes)
TB (Terabyte)  ≈ 1,000 GB  (≈ 1 Trillion Bytes)
```

## Why Size Matters:

- Determines device storage capacity and quota exhaustion.
- Dictates network transfer speed over campus connections.
- Determines if backups succeed or exceed mail attachment caps.

Even small percentage savings yield tens of gigabytes at scale.

EXERCISE & SOLUTION

# Viewing File Sizes in Your File Manager

### METHOD 1: PROPERTIES

Right-click folder > **Properties** (Win/Linux) or **Get Info** (Mac).

### METHOD 2: DETAILS VIEW

Switch File Explorer to **Details View** and sort by the **Size** column.

EXERCISE FINDING

### Largest File in Dataset

`big_file.data`

Sorting by size in Details view instantly reveals space hogs without opening individual items.

FUNDAMENTALS

# How File Compression Works

## Pattern Encoding

Uncompressed data often contains redundant repetitions:

```
AAAAAAABBBBBBBBBCCCCCCCC
```

Compressors encode repeated tokens mathematically (e.g. `7A 9B 8C`), dramatically shrinking required storage.

## Dual Advantages

1. **Shrinks disk footprint:** Fits within quotas and limits.
2. **Single-file bundling:** Packaging 5,000 small files into 1 archive avoids transfer overhead.

DATA INTEGRITY

# Lossless vs. Lossy Compression

## Lossless (Standard for Research)

Preserves 100% of bit-level data: `Original == Restored`.

**Formats: ZIP · GZIP · 7Z · PNG · FLAC**

Essential for numerical, tabular, code, and scientific measurements.

## Lossy (Caution in Research)

Achieves high ratio by permanently discarding fine information.

**Formats: JPEG · MP3 · MP4**

Risk: May destroy pixel-level quantitative features in microscopy or sensor audio.

OPERATING SYSTEM TOOLS

# How to Compress a Folder

### Windows

Right-click folder >

**Send To > Compressed (zipped) Folder**

Creates `archive.zip` in place.

### macOS

Right-click folder >

**Compress "folder_name"**

Generates standard ZIP archive.

### Linux

Right-click folder >

**Compress...**

Supports .zip, .tar.gz, .7z.

BEST PRACTICE

# When Should You Compress?

### Ideal For Compression

- Uncompressed plain text, CSVs, logs, TXT
- Inactive project archives (e.g. `archive.zip`)
- Preparing multi-file bundles for transfer

### Avoid or Ineffective

- Actively edited or frequently queried data
- Formats already compressed: JPG, PNG, MP4, PDF
- Re-zipping pre-compressed files saves ~0% space!

PRACTICAL EXERCISE

# Challenge: Measuring Compression Gains

**Determine the size of the cleaned project folder, then:**

1. Create a ZIP archive of the dataset.
2. Measure the size of the newly created ZIP file.
3. Calculate the percentage reduction in size.

**Discussion:** Was the reduction larger or smaller than expected? Why?

RESULTS & ANALYSIS

# Compression Ratio Depends on Data Type

### High Compression (50%–90%)

- Plain text files (`.txt`, `.csv`, `.tsv`)
- Raw unformatted database logs
- Bitmap uncompressed graphics (`.bmp`)

### Negligible Compression (0%–5%)

- JPEG photos, MP3/WAV audio
- PNG charts and PDF manuscripts
- Modern Office docs (`.docx`, `.xlsx` are already ZIP containers!)

FORMAT SELECTION

# File Formats Impact Storage Directly

### Bitmap vs. PNG Example

**sample.bmp:** Stores every single pixel colour value explicitly → Very large.

**sample.png:** Built-in lossless deflation algorithm → Fraction of the size with identical visual fidelity.

### Text vs. Binary Tables

**CSV:** Universal, transparent, and portable across all tools.

**Parquet / HDF5:** Compact binary storage; orders of magnitude faster read/write on huge datasets.

DISCUSSION 1

# How Have You Shared Data Before?

## Think about the last time you transferred research data to a colleague:

**CHANNELS USED**

Email · OneDrive · USB · SharePoint · Cloud

**OBSTACLES FACED**

File limits · Broken links · Corrupted syncs

EVALUATION

# Transfer Options: Small to Medium Scale

### Email Attachments (< 25 MB)

• Simple, universal for standalone memos and tiny scripts.

Disadvantages: Spawns unmanaged duplicate copies; insecure for personal data.

### OneDrive / SharePoint Links

• Single source of truth without sending physical copies.

Advantages: Share permissions can be revoked if sent in error.

LARGE SCALE

# Enterprise & HPC Transfer Services

### Globus (Gigabytes to Terabytes)

• High-performance, fault-tolerant grid transfer for HPC.

Recovers automatically from interrupted connections.

### Institutional Drop-off (e.g. dropoff.ncl.ac.uk)

• Encrypted temporary staging for medium datasets.

Features download receipts and automated expiration.

SECURITY HYGIENE

# Sharing Confidential Data & Sanitisation

### Email Is Never Secure

Emails pass unencrypted through dozens of intermediate relay servers.

**Never send sensitive participant data or passwords via email attachments.**

### Data Sanitisation Checklist

- Scrub autogenerated author names & track changes
- Transmit decryption passwords over a separate channel (SMS/phone)
- Prefer link sharing where permissions can be revoked if mistyped

SCENARIO MATCHING

# Challenge: Choosing A Transfer Method

**Match the transfer tool to the scenario with your partner:**

1. A 5 MB summary document for an external collaborator
2. A 300 MB tabular dataset shared among a research group
3. A 2 TB raw imaging dataset destined for HPC simulation nodes
4. Final cleaned dataset accompanying a published journal article

RECOMMENDED CHANNELS

# Recommended Transfer Matches

1. **5 MB document:** Email attachment or direct OneDrive link.
2. **300 MB group data:** OneDrive / SharePoint shared link, or institutional File Drop-off.
3. **2 TB HPC dataset:** Globus endpoint or high-speed rsync over SSH.
4. **Published journal output:** Public repository (Zenodo / Figshare) with a permanent DOI.

SUMMARY

# Episode 6 Key Points

**Measure First:** Use Details view to pinpoint large files before deciding on action.

**Lossless for Science:** ZIP preserves bit-for-bit data integrity; avoid lossy compression.

**Pre-compressed Formats:** Re-zipping PNGs or PDFs yields near-zero space reduction.

**Scale Matching:** Match tools (email < cloud links < Drop-off < Globus) to dataset scale.

**The Invariant:** Sanitize metadata and never send unencrypted confidential data over email.

LOOKING AHEAD

# Next: Episode 7

We now know how to measure, compress, and transfer data efficiently.
But what data should actually be kept, archived, or safely purged?

**Data Lifecycles and Retention**